

GILBERT YIGA.

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Professional Summary

Machine Learning Engineer with 2+ years of experience building and deploying production-scale AI/ML solutions. Specialized in Natural Language Processing for low-resource African languages, with proven expertise in end-to-end ML pipeline development—from data preparation and feature engineering through model deployment and monitoring. Published researcher with a strong foundation in working with large-scale datasets, developing multilingual language models, and implementing MLOps practices. Experienced in Python, SQL, and ML frameworks (TensorFlow, PyTorch, Scikit-learn) with hands-on production deployment experience. Eager to expand expertise in data lake analytics and big data engineering.

Education

- M.Sc. in Computer Science (Ongoing) - Makerere University, Kampala
- B.Sc. in Computer Science (First Class - Best Student Of The Class) - Makerere University.

Publications

- Sunflower: Expanding African Language Coverage in LLMs - arXiv, 2025
- Speech Data Requirements for ASR in African Languages - arXiv, 2025
- Real-Time Spoken Instruction & Translation in Ugandan Languages - OpenReview, 2024
- Explainable AI for Black Sigatoka Detection - Springer, 2023
- Transparent Intelligent Vision for Black Sigatoka Detection - Springer, 2023

Professional Experience

Graduate Researcher - Makerere University, Department of Computer Science (Present)

- Developing an end-to-end Speech-to-Speech Translation system for Luganda, addressing critical challenges in low-resource language AI
- Investigating safety considerations in translation systems: bias detection, evaluation frameworks for high-stakes contexts (medical, legal), and alignment in cross-cultural communication
- Implementing transformer-based architectures and adapting state-of-the-art models (Wav2Vec, Whisper) for low-resource settings
- Contributing to democratizing access to AI technologies for East African languages
- Collaborating with linguists and domain experts to ensure culturally appropriate and safe AI systems

Part-time Software Engineer - Sunbird AI (Present)

- Designed and deployed production ML systems serving real-time translation and speech inference with high availability
- Built end-to-end data pipelines for collection, preprocessing, validation, model training, evaluation, and deployment
- Developed monitoring infrastructure tracking system performance, reliability, usage metrics, and evaluation indicators
- Designed structured data collection systems using Django webhooks integrated with WhatsApp Business API
- Implemented evaluation frameworks using BLEU, chrF, and COMET to ensure measurable performance benchmarking
- Collaborated with research and engineering teams to align technical outputs with institutional and public service use cases.

Part-time Machine Learning Engineer

Emergent AI - April 2024 - Feb 2025

- Applied machine learning and computer vision techniques to health-sector datasets
- Built structured feature extraction pipelines for large-scale medical imaging data
- Designed model validation and performance reporting workflows to support data-informed health insights
- Translated stakeholder requirements into measurable technical performance metrics

Software Developer

Pahappa Ltd, Ntinda, Muteesa II - August 2022 - October 2023

- Developed enterprise CRM systems supporting business operations at scale
- Designed relational database schemas and optimized SQL queries for reporting efficiency
- Integrated REST APIs across systems to improve interoperability
- Contributed to digital workflow optimization initiatives across client organizations

Machine Learning Engineer Intern

AI Lab, Makerere University - November 2021 - May 2022

- Developed predictive computer vision models for agricultural disease detection
- Processed and analyzed large agricultural image datasets
- Contributed to research on applied AI systems for sustainable development

Public Projects

Production ML Model Deployment

- Deployed Sunbird translation and speech recognition models to production with high availability
- Implemented model serving infrastructure handling real-time inference requests
- Built monitoring and logging systems to track model performance and usage metrics

WhatsApp API Integration & Data Collection System

- Designed a webhook architecture using Django for handling real-time messaging
- Built an admin review system for data management and quality control
- Integrated official WhatsApp Business API for scalable messaging solutions

Worked on the Sunbird translation and speech transcription systems.

- These are React applications that are public on the Sunbird GitHub repo.

Model deployment with the Sunbird models.

- Deployed both the Sunbird translation and Speech recognition models to production.

Low-resource Language Identification.

- Developed a language identification model with text for multiple local languages.

Blood Donation Uganda.

- Developed a mobile application to facilitate blood donations across Uganda.

Machine Learning for Black Sigatoka Detection.

- Created a model to detect Black Sigatoka in banana plantations, implemented in mobile applications.

Technical Skills

Programming & Database:

Python (Advanced), SQL (Proficient), JavaScript/TypeScript, Java

Machine Learning Frameworks:

TensorFlow, PyTorch, Scikit-learn, XGBoost, Transformers, Pandas, NumPy

ML Specializations:

Natural Language Processing, Computer Vision, Machine Translation, Speech Recognition, Predictive Modeling, Feature Engineering, Model Selection & Optimization

MLOps & Production Deployment:

Production ML pipelines, Model versioning, Performance monitoring, API integration, Docker, Git, CI/CD practices, Distributed training (DeepSpeed, FSDP)

Development Tools & Frameworks:

Django, Flask, React, Node.js, REST APIs, Git, Docker

Data Engineering & Analytics (Growing Proficiency):

SQL query optimization, Data pipeline design, Large-scale data processing, Statistical analysis

Research Methods:

Experiment tracking, Model evaluation metrics (BLEU, chrF, COMET), Technical documentation, Statistical analysis

Certifications

- Mobile Development with Flutter, Online | 2022
- Web Development, Online | 2021
- Text Classification NLP, Online | 2024
- Machine Learning in Production, Online | 2024

- Human Research Protection Training by US Department of Health and Human Services (HHS), Online | 2024

Languages

- English, and Luganda

Interests

- NLP, Machine Learning, Data Analysis, Data Science, and Software Development.

Referees

Available on request.